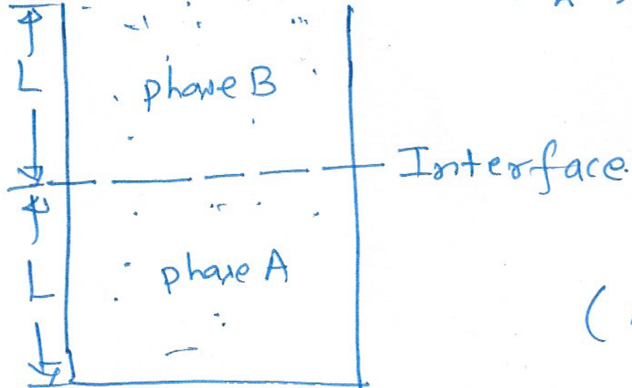
① Consider two immiscible liquids A & B containing a common solute with its intinial concentration C_{A0} & C_{B0} respectively. Both phases A & B (containing solute) are brought in contant in a container, as shown below.

D_A & D_B are solute diffusivities in liq A & B respectively.



(Assume Fick's law is valid & interface is always at equilibrium, $C_B = mC_A$)

Find out the mass transfer rate (time-dependent) at interface (for $t << L^2/D_A$, $t << \frac{L^2}{D_B}$)

$$\left[\ \frac{d(\mathrm{erf}(x))}{dx} = \frac{2}{\sqrt{\pi}}\exp(-x^2)\ \right]$$